Math Hard Question 11

1 ★ Mark for Review ABC

In the given equation, a and b are positive constants. The sum of the solutions to the given equation is $k(6a + b)$, where k is constant. What is the value of k?

$$24x^2 - (12a + 2b)x + ab = 0$$

A. $\frac{1}{12}$

B. $-\frac{1}{12}$

C. 12

D. -12

2 ★ Mark for Review ABC

The positive number a is 2136% of the sum of the positive numbers b and c, and b is 89% of c. What percent of b is a?

A. 4536%

B. 2400%

C. 4037%

D. 2225%

3 ★ Mark for Review ABC

The graphs in the xy-plane of the following quadratic equations each have x-intercepts of -2 and 4. The graph of which equation has its vertex **farthest from the x-axis**?

A. $y = -7(x+2)(x-4)$

B. $y = \frac{1}{10}(x+2)(x-4)$

C. $y = -\frac{1}{2}(x+2)(x-4)$

D. $y = 5(x+2)(x-4)$

4 ★ Mark for Review ABC

In one hour, the distance, $d(s)$, in kilometers that a ferry can travel up and down a river flowing with a constant speed s, in meters per second, is:

$$d(s) = 10.7 - 1.2s^2$$

where s and $d(s)$ are positive.

Which of the following equivalent expressions for $d(s)$ contains the speed of the river in meters per second, as a constant or coefficient, for which the ferry can travel a distance of 8 kilometers in one hour?

A. $8 - 1.2(s-1.5)(s+1.5)$

B. $8 + 1.2(2.25 - s^2)$

C. $8(1 - 0.15s^2) + 2.7$

D. $11.9 - 1.2(s-1)^2 - 2.4s$

5 ★ Mark for Review **ABC**

A minor league hockey team has been collecting ticket sales data over the past year. At a current price of \$25 per ticket, an average of 4000 seats are purchased. They predict that for each \$1 increase in ticket price, 100 fewer tickets will be sold.

Which of the following functions best models the amount of money, y, that the hockey team expects to collect from ticket sales based on an x-dollar increase in ticket price?

A. $y = (25 + x)(4000 - 100x)$

B. $y = (25 - x)(4000 + 100x)$

C. $y = x(4000 - 100x)$

D. $y = 4000(25 + x)$

6 ★ Mark for Review **ABC**

In the given expression:

$$34z^{14} + bz^7 + 70,$$

b is a positive integer. If $qz^7 + r$ is a factor of the expression, where q and r are positive integers, what is the greatest possible value of b?

7 ★ Mark for Review ABC

Point N lies on a unit circle in the xy-plane and has coordinates $(1, 0)$. Point O is the center of the circle and has coordinates $(0, 0)$. Point M also lies on the unit circle, and the measure of angle $\angle NOM$ is $\frac{266\pi}{3}$ radians. If the coordinates of point M are (a, b), where a and b are constants, what is the value of a?

A. $\frac{1}{2}$

B. $\frac{\sqrt{3}}{2}$

C. $-\frac{1}{2}$

D. $-\frac{\sqrt{3}}{2}$

8 ★ Mark for Review ABC

For each of two rectangles, the ratio of the rectangle's length to its width is $10 : 5$. The width of rectangle X is 16.5 times the width of rectangle Y. How does the length of rectangle X compare to the length of rectangle Y?

A. The length of rectangle X is 0.5 times the length of rectangle Y

B. The length of rectangle X is 2 times the length of rectangle Y

C. The length of rectangle X is 16.5 times the length of rectangle Y

D. The length of rectangle X is 33 times the length of rectangle Y

9 ★ Mark for Review ABC

In the xy-plane, a circle has center C with coordinates (h, k). Points A and B lie on the circle. Point A has coordinates $(h+1, k+\sqrt{102})$, and $\angle ACB$ is a right angle. What is the length of $\overline{AB}$?

A. $\sqrt{206}$

B. $2\sqrt{102}$

C. $103\sqrt{2}$

D. $103\sqrt{3}$

10 ★ Mark for Review ABC

$$y = a * (\frac{6}{b})^x - c$$

How many times does the graph of the given equation in the xy-plane cross the x-axis, where a, b, and c are positive constants such that $a > 6$ and $b > c$?

A. Zero

B. One

C. Two

D. Three

11 ★ Mark for Review ABC

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $9x - 110°$. Which of the following could **NOT** be the sum of the measures of any two of these angles?

A. $-18x + 220°$

B. $-18x + 580°$

C. $18x - 220°$

D. $180°$

12 ★ Mark for Review ABC

The function f is defined by:

$$f(x) = a(3.1^x + 3.1^b),$$

where a and b are integer constants and $0 < a < b$. The functions g and h are equivalent to function f, where k and m are constants.

Which of the following equations displays the **y-coordinate of the y-intercept** of the graph of $y = f(x)$ in the xy-plane as a constant or coefficient?

I. $g(x) = a(3.1^x + k)$

II. $h(x) = a(3.1^x) + m$

A. I only

B. II only

C. I and II only

D. Neither I nor II